Communicating with Data – Visualization Literacy

Dr. Ab Mosca (they/them)

Plan for Today

- Define Visualization Literacy
  - As a set of skills
  - As a model / concept
- Testing Visualization Literacy
- Barriers to Visualization Literacy

Brainstorm

- What is visualization literacy?
- How can we measure it?
- Why is it useful?

Exploring Definitions– Visualization Literacy as a Skillset

No agreed upon definition of visualization literacy. Many conceptualizations exist

- The next 10 slides present proposed definitions for visualization literacy through skill conceptualizations.
- Make 10 groups and count off, read the definition numbered the same as your group number.
- Pair up: Groups `1&6, 2&7, 3&8, 4&9, 5&10`
  - What are the similarities and differences between your definitions?
- Pair up: Groups `1&6&5, 2&7&4, 3&8&9&10`
  - What are the similarities and differences between your definitions?
- Whole class: What are the similarities and differences between your definitions?
- What aspects of these definitions do we like? Which do we disagree with?

Skill Definitions (1)

Vermeersch & Vandenbroucke define visualization literacy as:

- **Perception**
  - cultivating a "good eye" in experiencing the visual
  - being curious, sensitive, focused, selective, and deriving pleasure from it
- **Imagination and creation, Conceptualization**
  - Using perceived images to generate something new, based on the capacity to understand image structure (ex. points, lines, angles, open and closed figures)
  - Transforming concrete perceptions into abstract representations
- **Analysis of images**
  - identifying structures or patterns within a visualization

S. Beschi, D. Falessi, S. Golia and A. Locoro, "Characterizing Data Visualization Literacy for Standardization: A Systematic Literature Review," in *IEEE Access*, vol. 13, pp. 65704-65725, 2025, doi: 10.1109/ACCESS.2025.3559298.

Skill Definitions (2)

Kendra defines visualization literacy as:

- **Visual reading**
  - Ability to interpret, analyze, understand, evaluate, and translate visualizations
- **Visual writing**
  - Ability to create a produce visualizations
- **Other visual literacy skills**
  - Ability to think visually and learn visually

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Skill Definitions (3)

Rybarczyk et al. define visualization literacy as:

- **Basic**
  - Ability to identify patterns and trends in data
  - Ability to connect source of data
- **Advanced**
  - Ability to synthesize conclusions from data
  - Ability to propose follow up experiments and predict results

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Skill Definitions (4)

Arneson and Offerdahl et al. define visualization literacy as:

- **Knowledge**
  - Ability to memorize, recognize, recall, and retrieve data
- **Comprehension**
  - Ability to understand, interpret, infer, exemplify, and classify
- **Application**
  - Ability to execute, implement, and apply visualization
- **Analysis**
  - Ability to differentiate, discriminate, organize, integrate, and deconstruct visualization
- **Evaluation**
  - Ability to check, critique, judge, and test visualization
- **Synthesis**
  - Ability to generate, hypothesize, plan, design, construct, reorganize, and produce visualization

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Skill Definitions (5)

Arneson and Offerdahl et al. define visualization literacy as:

- **Knowledge**
  - Ability to memorize, recognize, recall, and retrieve data
- **Comprehension**
  - Ability to understand, interpret, infer, exemplify, and classify
- **Application**
  - Ability to execute, implement, and apply visualization
- **Analysis**
  - Ability to differentiate, discriminate, organize, integrate, and deconstruct visualization
- **Evaluation**
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Skill Definitions (6)

Kalaf-Hughes defines visualization literacy as:

- **Information literacy**
  - Ability to retrieve and organize information
- **Visual literacy**
  - Ability to perceive, analyze, represent, and use multi-modality
- **Critical thinking**
  - Ability to think analytically for problem solving and decision making
- **Communication**
  - Ability to communicate effectively, collaborate with peers, speak publicly
- **Adaptability and lifelong learning**
  - Ability to adapt and learn for life

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Skill Definitions (7)

Mnguni et al. define visualization literacy as:

- **Internalization**
  - Ability to arrange, order, organize, classify, reorganize, find, locate, focus, ground, and perceive visual stimuli such as luminance, color, and motion
- **Conceptualization**
  - Ability to analyze, interpret, asses, evaluate, examine, investigate, compare, relate, critique, imagine, describe, discuss, explain, discriminate, judge, manipulate objects using geometrical properties and operations, recall, and retrieve
- **Externalization**
  - Ability to complete, illustrate, infer, predict, outline, propose, develop, formulate, devise, construct, create, produce, invent, use, and define the need for an image

S. Beschi, D. Falessi, S. Golia and A. Locoro, "Characterizing Data Visualization Literacy for Standardization: A Systematic Literature Review," in *IEEE Access*, vol. 13, pp. 65704-65725, 2025, doi: 10.1109/ACCESS.2025.3559298.

Skill Definitions (8)

Reddy et al. define visualization literacy as:

- the ability to find images, interpret and analyze images, evaluate images, use images effectively, create visual medial, and use images ethically citing visuals

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Skill Definitions (9)

Krejci et al. define visualization literacy as:

- **Elementary**
  - Ability to interpret visualization
- **Intermediate**
  - Ability to reason, explain, and evaluate visualization
- **Advanced**
  - Ability to synthesize, reflect, and judge visualization

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Skill Definitions (10)

Hedayati and Kay define visualization literacy as:

- the ability to deconstruct visualization, and design empathy for visualization

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Exploring Definitions– Visualization Literacy as a Model / Construct

No agreed upon definition of visualization literacy. Many conceptualizations exist

- The next 7 slides present proposed definitions for visualization literacy through models or constructs.
- Make 7 groups and count off, read the definition numbered the same as your group number.
- Pair up: Groups 1&7, 2&6, 3&5&4
  - What are the similarities and differences between your definitions?
- Pair up: Groups 1&7&2, 3&5&4&6
  - What are the similarities and differences between your definitions?
- Whole class: What are the similarities and differences between your definitions?
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Model/ Construct Definitions (1)

Borner et al. define visualization literacy as:

- **DVL-Framework typology**
  - Defines core concepts of visualization: insights, data scales, kinds of analyses, kind of visualization, graphic symbols, encoding variables, interactions
- **DVL-Framework process**
  - Defines steps for constructing and interpreting a visualization: identification of stakeholders' insights, relevant datasets, data pre-processing and visualization, deployment, and interpretation of results

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Model/ Construct Definitions (2)

Bresciani and Eppler define visualization literacy as:

- **Potential pitfalls**
  - Poor encoding (designer errors) through—
    - ambiguity, breaking conventions, cryptic encoding, hiding/obscuring, over-complexity, over-simplification, redundancy, task-visualization fit, unclear, etc.
    - disturbing or boring content, ugly, wrong use of color
    - exercise of power, inhibit conversation, unequal participation
  - Poor decoding (user errors) through–
    - channel thinking, poor perceptual skills, focus on low-relevance items, high reliance on training and resources, misuse, overload
    - visual stress, personal likes and dislikes, prior knowledge and experience
    - altered behavior, cultural and cross-cultural differences, recency effect, hiding differences of opinion

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Model/ Construct Definitions (3)

Davis et al. define visualization literacy as:

- **Visualization tasks**
  - Estimate and compare values
  - judge correlations
  - identify trends and outliers
  - spot suitability

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Model/ Construct Definitions (4)

Galesic and Garcia-Retamero define visualization literacy as:

- **Graph literacy levels**
  - Level 1: Read the data
  - Level 2: Read between the data
  - Level 3: Read beyond the data

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Model/ Construct Definitions (5)

Lee et al. define visualization literacy as:

- **Cognitive characteristics**
  - Numeracy
  - need for cognition
  - visualizer-verbalizer capacity

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Model/ Construct Definitions (6)

Locoro et al. define visualization literacy as:

- **Construct levels of VIL**
  - Map
  - Integrate
  - Compute
  - Reason
  - Infer
  - Explain

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Model/ Construct Definitions (7)

Lundgard and Satyanarayan define visualization literacy as:

- **Model levels of DVL**
  - Enumerating visualization construction properties
  - Reporting statistical concepts and relations
  - Identifying perceptual and cognitive phenomena
  - Elucidating domain specific insights

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Brainstorm

Those are a LOT of different ways to think about visualization literacy.

If you think about assessing visualization literacy, do any of those definitions lend themselves well to creating an assessment? Which ones? Why?

Brainstorm

Choose one definition. Outline a test, based on the definition you chose, to determine someone's visualization literacy level.

Assessing Visualization Literacy

- Most widely used assessment is VLAT (Visualization Literacy Assessment Test)
- VLAT includes 12 data visualizations and 53. multiple choice questions
- Demo: https://revisit.dev/study/library-vlat/SURHUmJIbl03Uk4xUUhNUHQ1VTRIQT09
  - Click "Next" at the bottom of the page to take the assessment

S. Lee, S.-H. Kim, and B. C. Kwon, "VLAT: Development of a Visualization Literacy Assessment Test," IEEE Transactions on Visualization and Computer Graphics, vol. 23, no. 1, pp. 551-560, Jan. 2017

Assessing Visualization Literacy

- There is also a minVLAT!
- Much, much shorter than the long-form VLAT
  - 12 items instead of 53
- Demo: https://tools.visualdata.wustl.edu/experiment/#/
  - Click "I agree to participate" at the bottom of the page to take the assessment

Pandey, S. and Ottley, A. (2023), Mini-VLAT: A Short and Effective Measure of Visualization Literacy. Computer Graphics Forum, 42: 1-11. https://doi.org/10.1111/cgf.14809

Barriers to Visualization Literacy

- Answering a question with data visualization can be thought of as 5 parts:
  1. Posing the question
  2. Translating the question into a visual query
  3. Grasping the visualization's layout and visual encoding
  4. Decoding the relevant values
  5. Synthesizing an answer

Carolina Nobreet al. 2024. Reading Between the Pixels: Investigating the Barriers to Visualization Literacy. In Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 197, 1–17. https://doi.org/10.1145/3613904.3642760

Barriers to Visualization Literacy

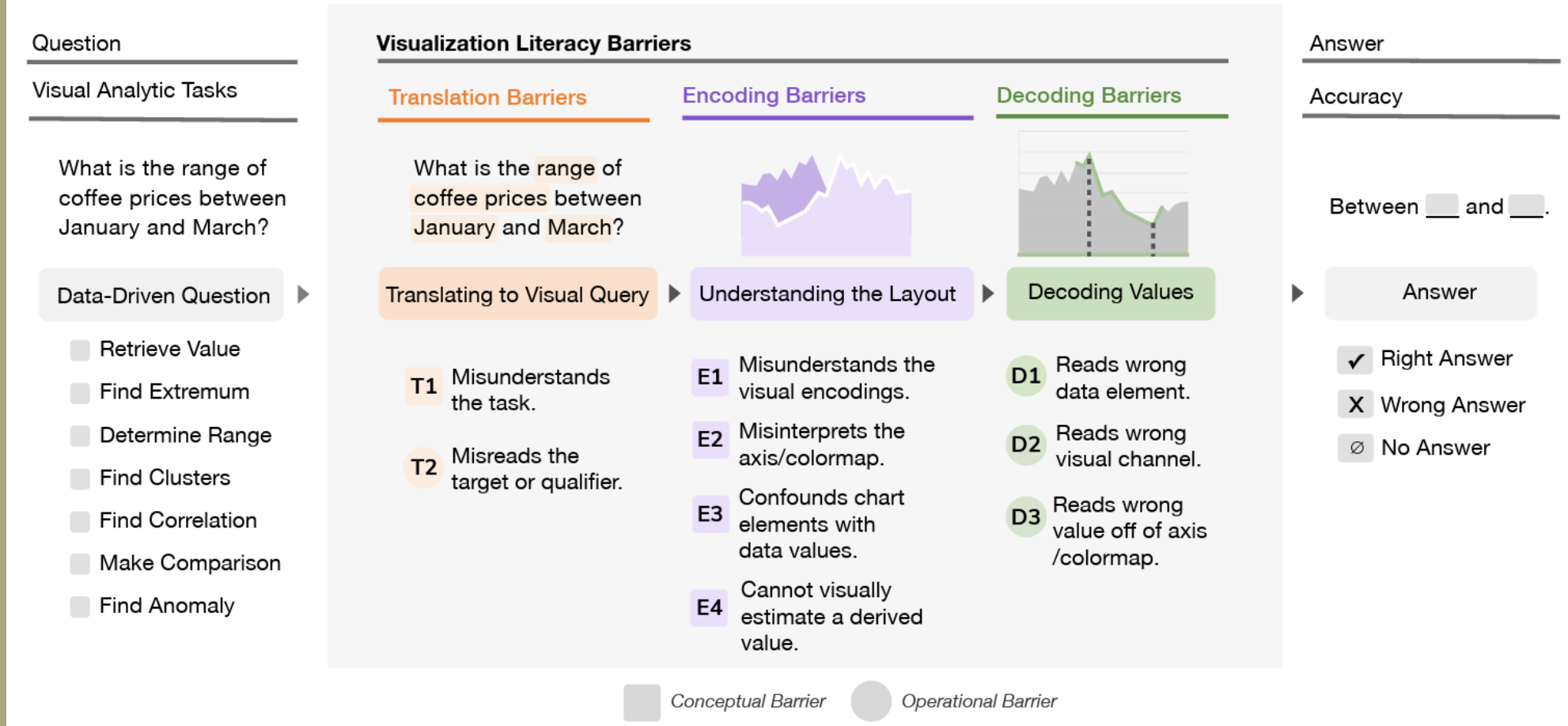
- Answering a question with data visualization can be thought of as 5 parts:
  1. Posing the question
  2. Translating the question into a visual query
  3. Grasping the visualization's layout and visual encoding
  4. Decoding the relevant values
  5. Synthesizing an answer

  Barriers in these stages are either (a) ***translation***, (b) ***encoding***, or (c) ***decoding*** barriers.

  Some barriers are ***conceptual*** (resulting from the viewer's understanding of the task of visualization), others are ***operational*** (resulting from misreading the question or elements in a visualization)

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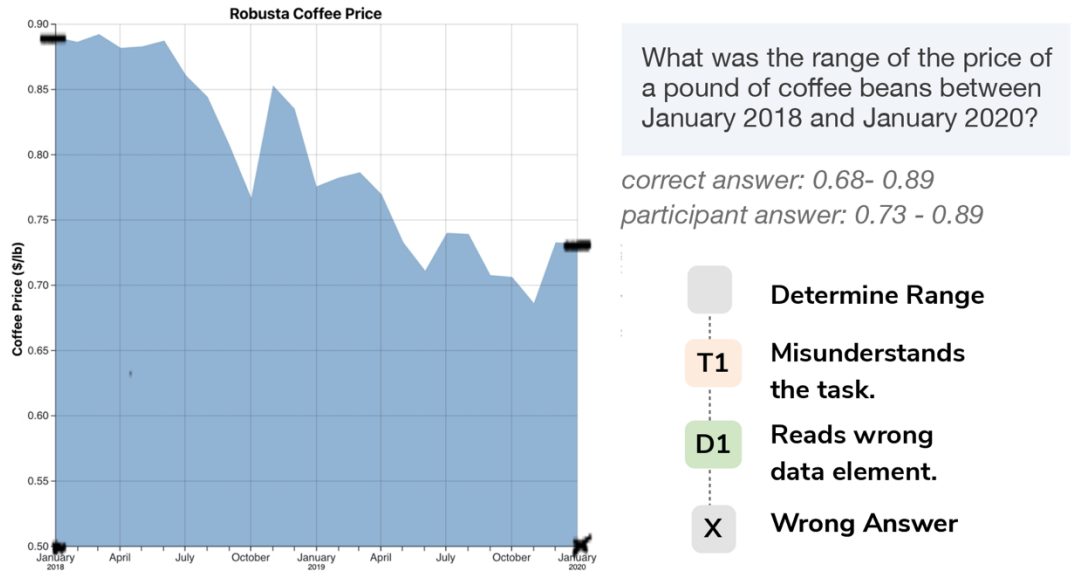


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Barriers to Visualization Literacy

Translation barrier – misreads task

Decoding barrier – reads wrong element



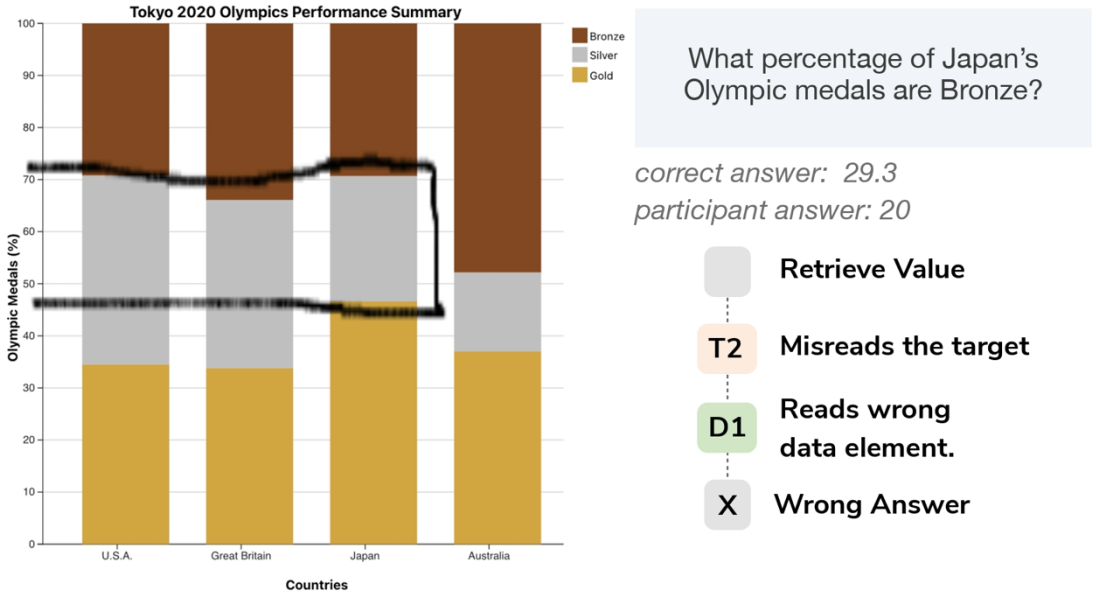
(a) Example of a *T1: Misunderstands the task* mistake. The participant misunderstood what "range' of the data meant and instead read the values only at the start and end dates. This mistake also led to D1 mistake, where the participant read off the wrong data element in the chart.

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Barriers to Visualization Literacy

Translation barrier– misreads target

Decoding barrier – reads wrong data element



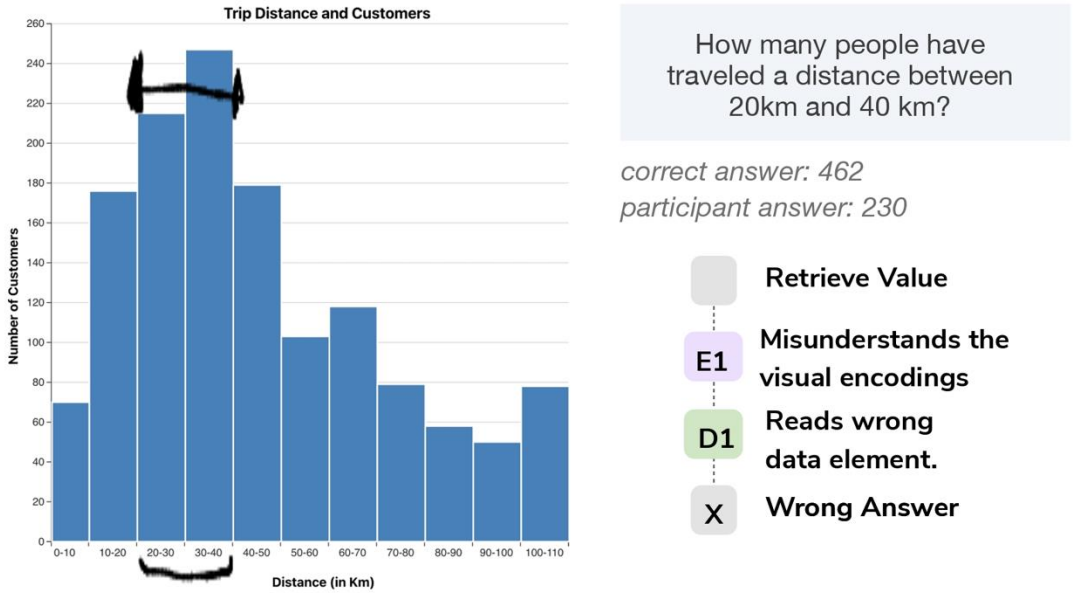
(b) Example of a *T2: Misreads the target/qualifier mistake.* The participant looked up the silver medals for Japan instead of the Bronze as asked in the task. This mistake also cascaded into a D1 decoding mistake, where the participant read off the wrong data element from the chart.

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Barriers to Visualization Literacy

Encoding barrier—misunderstands visual encoding

Decoding barrier–reads wrong value



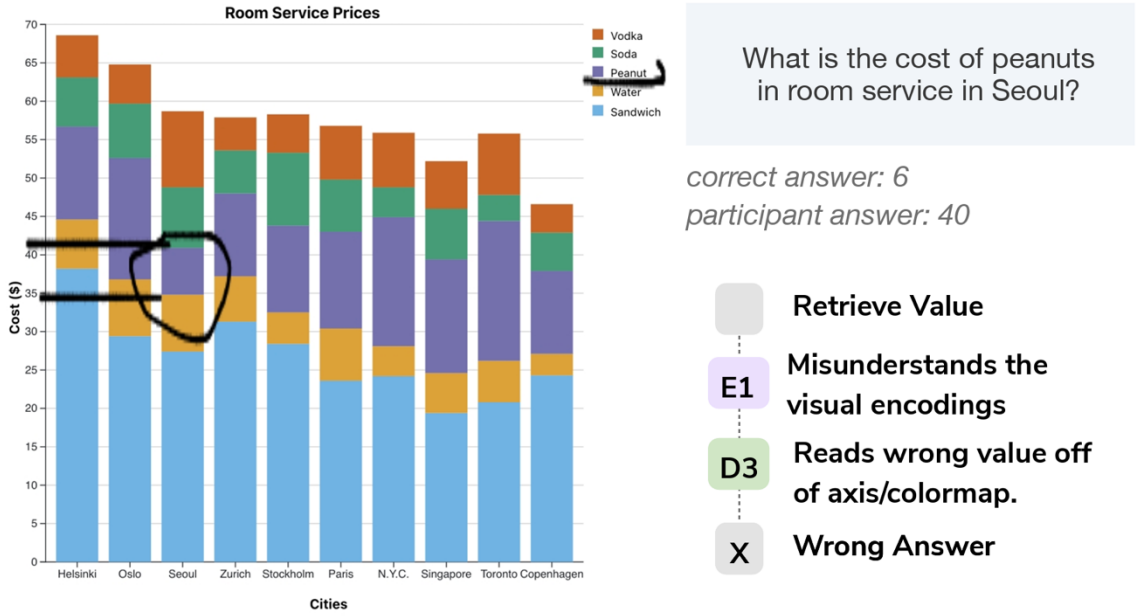
(a) Example of a *E1: Misunderstands the visual encodings* mistake. The participant misunderstood what each bar in the histogram represents. This also led to a D1 decoding mistake, where they read off the values for the wrong bars in the chart.

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Barriers to Visualization Literacy

Encoding barrier– misunderstands visual encoding

Decoding barrier— reads wrong bars



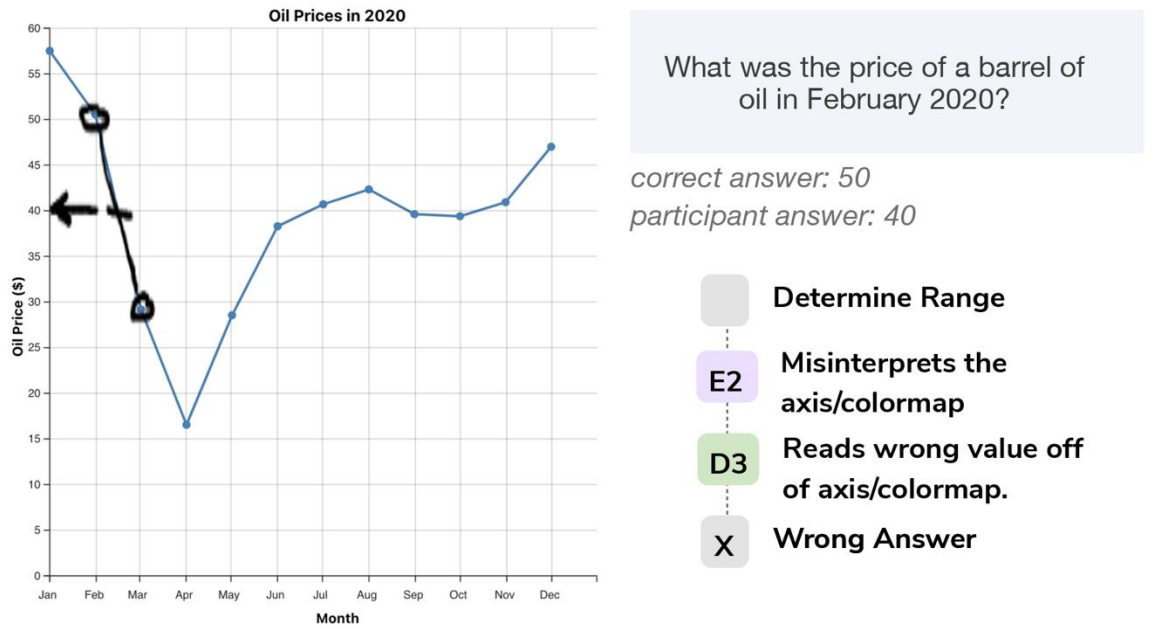
(b) Example of a *E1: Misunderstands the visual encodings* mistake. The participant interpreted stacked bars as representing the min-max range for each category. This conceptual hurdle led to a subsequence D3 mistake, where the participant read off the wrong value from the axis (the middle of the bar).

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Barriers to Visualization Literacy

Encoding barrier—misinterprets axis or colormap

Decoding barrier–reads wrong value



(a) Example of a *E2: Misinterprets the axis or colormap* mistake. The participant interprets the line connecting the Feb and March data points as the days in February. This conceptual flaw also led to the D3 mistake of reading the wrong value off of the Y axis.

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Barriers to Visualization Literacy

Encoding barrier—misinterprets axis or colormap

Decoding barrier—reads wrong value

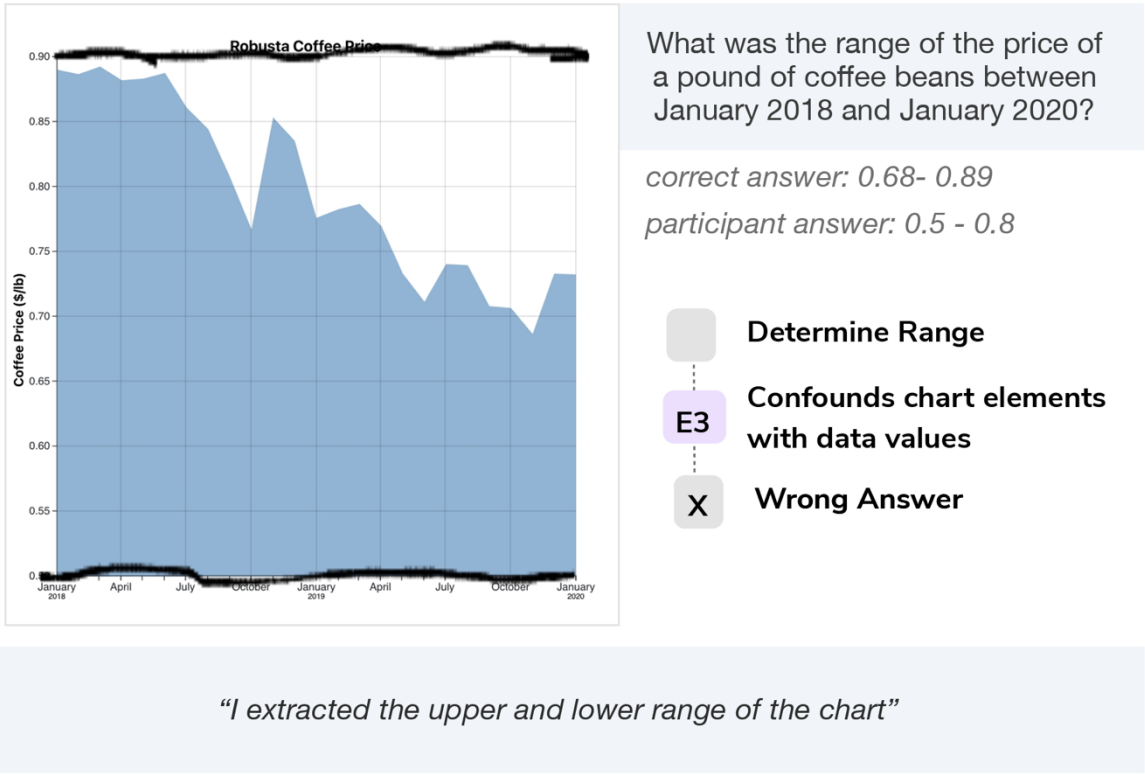


(b) Example of a *E2: Misinterprets the axis or colormap* mistake. The participant correctly identifies the color but does not realize that the color refers to the range 6- 8%. As a result, they also fall victim to the D3 mistake of reading off the wrong value from the colormap.

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Barriers to Visualization Literacy

Encoding barrier—confounds chart and data values

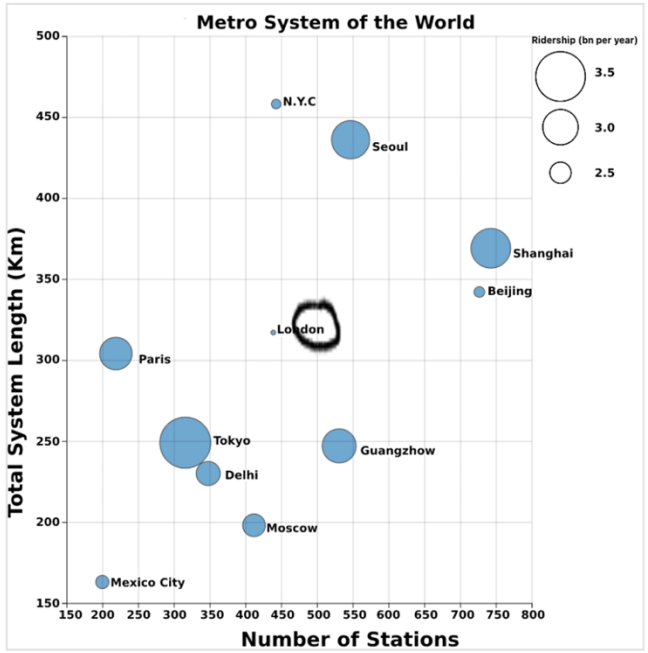


(a) Example of an *E3: Confounds chart elements with data values* mistake. This participant read the min and max values of the axis as the min and max values in the dataset.

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Barriers to Visualization Literacy

Encoding barrier—confounds chart and data values



The average metro system in this chart has approximately 300 stations and is 200km in length. True or False?

correct answer: False
participant answer: False

Find Clusters

E3 Confounds chart elements with data values

X Wrong Answer

"The area I marked is the centre of the chart. It represents the average metro system of the world."

(b) Example of an *E3: Confounds chart elements with data values* mistake. This participant estimated the center of the chart as the average value for the data.

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Your Turn!

- Sketch two visualizations
  - One should be designed for an audience with low visualization literacy
  - One should be designed for an audience with high visualization literacy
- Swap with someone
  - Without any hints, can you guess which visualization is for low visualization literacy, and which is for high visualization literacy?

Brainstorm

Is it possible to create a visualization that increases a reader's visualization literacy? Why or why not?

If you were forced to try, how would you design it? Why?